

Generate Power Using Microturbine

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Presented here is the circuit that generates power from a water pipe in a building using a micro-turbine. The generated electricity can be used to charge batteries, which can be used for emergency lightning or other such purposes.

Overhead water tanks are placed on the terrace of the house, and filled from the underground sump or borewell using an electric pump for domestic use. Water has high kinetic energy when it flows from a high to low level. Electricity can be generated using this kinetic energy with the help of a suitable microturbine. Based on the amount of waterflow in the pipe, power generated using the turbine can be varied.

Circuit and working

The circuit diagram for hydropower generation from a water pipe is shown in Fig. 1. It is built around six rectifier diodes 1N4007 (D1 through D6), DC-to-DC regulator LM2596 (IC1), Schottky diode 1N5822 (D7), inductor (L1), micro hydro-turbine (MHT) and a few other components.

The circuit is built around DC-to-DC step-down switching regulator IC LM2596. The three-phase outputs (RYB) of MHT are connected to CON1. The image of MHT used in this project is shown in Fig. 2.

Bridge rectifiers (comprising

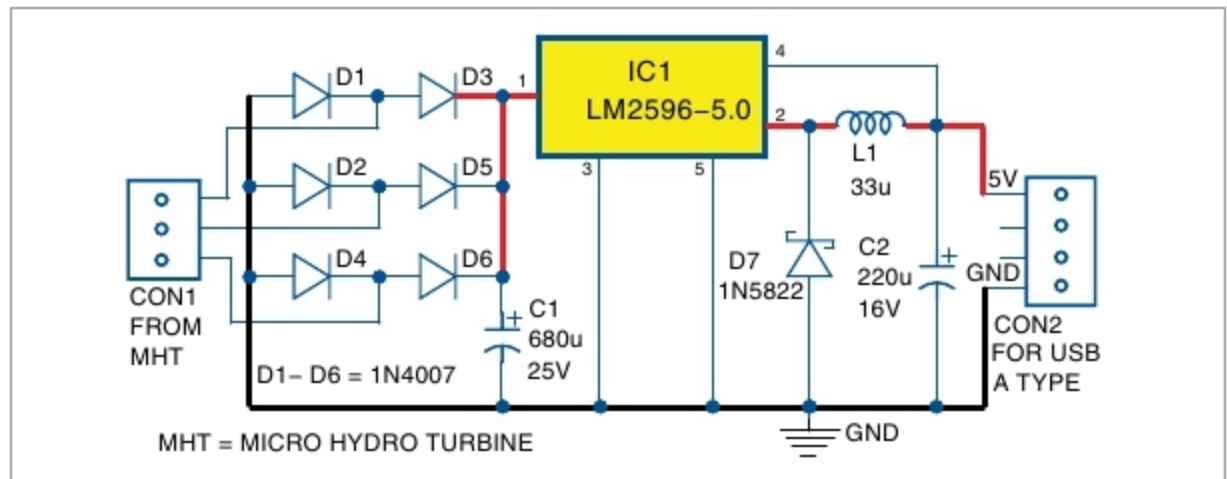


Fig. 1: Circuit diagram for hydropower generation



Fig. 2: A typical micro hydro-turbine (MHT)

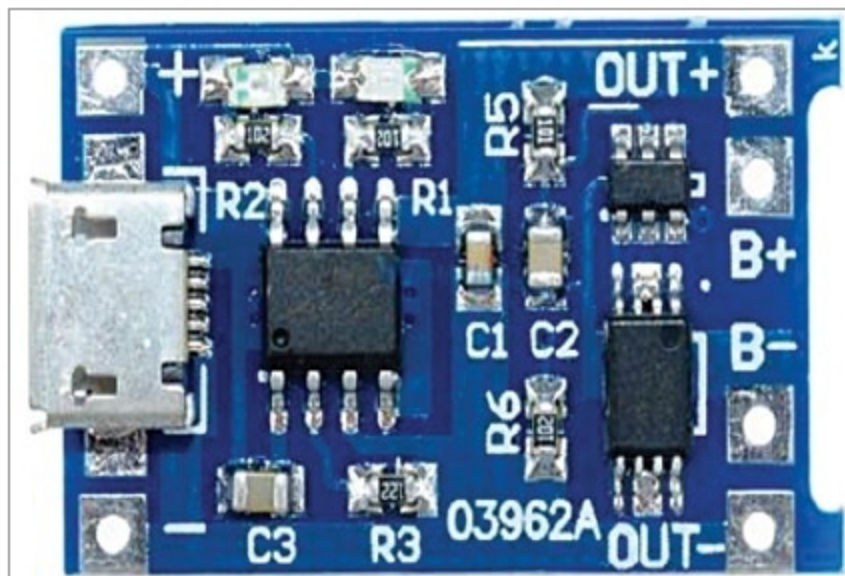


Fig. 3: TP4056 module

diodes D1 through D6) convert AC output from MHT to DC voltage. IC1, L1 and D7 perform the step-down

and an adjustable output version. For example, LM2596-5.0 gives 5V fixed output.

operation that converts high voltage to 5V DC. Capacitor C1 smoothens the rectified DC voltage, and C2 acts as a buffer for the output. 5V DC available at connector CON2 can be used to charge the single 18650 lithium-ion cell using TP4056 module shown in Fig. 3. An emergency light or some other gadgets can be powered by the single lithium-ion cell.

LM2596. LM2596 series of regulators are monolithic ICs that provide all active functions for a step-down (buck) switching regulator, which is capable of driving a 3A load with excellent line and load regulation. These devices are available in fixed output voltages of 3.3V, 5V, 12V

PARTS LIST

Semiconductors:

- IC1 - LM2596-5.0 DC-to-DC converter
- D1-D6 - 1N4007 rectifier diode
- D7 - 1N5822 Schottky diode

Capacitors:

- C1 - 680µF, 35V electrolytic
- C2 - 220µF, 16V electrolytic

Miscellaneous:

- CON1 - 3-pin connector
- CON2 - USB-A type
- Micro hydro-turbine (3-phase, 20V AC)
- Heat-sink for LM2596